

Wireless Inductive EV Charging Design

SAE J2954 WPT2 7.7 kW / 85 kHz link: coil design, SS and LCC-S compensation, losses and Biot-Savart field check

AT A GLANCE

7.7 kW / 85 kHz / 175 mm

Power / frequency / gap

214.8 / 175.8 uH

Coil L1 / L2 (with ferrite factor)

0.221 / 0.203

k nominal / +75 mm offset

97.8 %

Link efficiency, SS nominal

96.5 / 96.1 %

System DC-DC efficiency SS / LCC-S

5.4 uT (< 27 uT)

Field at person 1.0 m, 0.5 m

< 1 % error

Neumann M vs Maxwell exact

15 / 15

Tests re-run here

VERDICT

TESTS 15/15

RE-RUN HERE

DECLARED SUBSTITUTION

NO COMSOL / MAXWELL

ICNIRP 2010 PASS

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1. Objective and brief mapping

Brief: model an inductive charging system for EVs using COMSOL or ANSYS Maxwell, design primary and secondary coils, and analyse power-transfer efficiency and losses. I did not have COMSOL or Maxwell, so I replaced the finite-element step with an **analytical + numerical Python model**: published closed-form self-inductance (Mohan, Wheeler), a vectorised Neumann-integral filament solver for mutual inductance, a Biot-Savart field solver on a cut plane, and a phasor mesh model of the resonant link and SiC converters. Design case from SAE J2954: WPT2 7.7 kW to a 400 V battery, 85 kHz, Z2 gap class 140-210 mm (nominal 175), +/-75 mm X offset, ICNIRP 2010 27 uT limit. python main.py regenerates results/ in about 20 s.

2. Model and architecture

Primary: 20 turns, r_{in} 100 mm, r_{out} 300 mm, pitch 10.5 mm, 25.1 m of 400 x 0.1 mm litz (3.14 mm²). Secondary: 20 turns, 85-225 mm, 19.5 m. $L = \mu_0 N^2 d_{avg} c1/2 [\ln(c2/\rho_0) + c4 \rho_0^2]$ (Mohan); air-core 165.2 / 135.3 uH, design values with a declared ferrite factor $f_L = 1.30 \rightarrow 214.8 / 175.8$ uH (Wheeler +0.3 / +0.6 %, filament -1.9 / -2.9 %). $M = \mu_0/(4 \pi) \sum dL_i \cdot dL_j / |r_i - r_j|$ over 72 segments per turn, $x f_M = 1.40$ for ferrite. Litz R_{ac} at 85 kHz (skin depth 0.224 mm, proximity factor 1.63): 225 / 174 mOhm, 282 / 221 mOhm with capacitor ESR, Q 406 / 424.

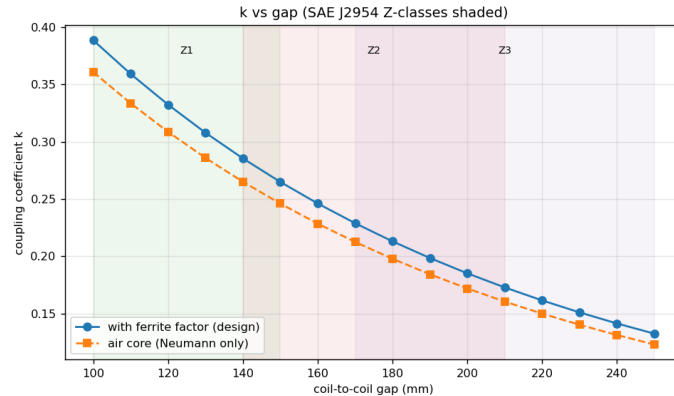


Fig. 1 - Coupling coefficient vs gap with the J2954 Z-classes shaded, air core vs ferrite-factor: $k = 0.286$ at 140 mm, 0.221 at 175 mm, 0.173 at 210 mm.

3. Method and implementation

■ SS compensation: $C1 = 1/(w_0^2 L1) = 16.32$ nF, $C2 = 19.94$ nF; reflected impedance $Z_r = w_0^2 M^2 / (R2 + R_L)$; $\eta_{max} = k^2 Q1 Q2 / (1 + \sqrt{1 + k^2 Q1 Q2})^2 = 97.8 \%$ at $k \sqrt{Q1 Q2} = 92$; $R_L = 8 V_{bat}^2 / (\pi^2 P) = 16.8$ ohm vs optimum 20.3 ohm. LCC-S: L_{f1} 42.5 uH, C_{f1} 82.5 nF, $C1$ 20.35 nF, load-independent $|I1| = |V1| / (w_0 L_{f1})$.

■ Circuit: 2 x 2 (SS) or 3 x 3 (LCC-S) mesh phasor system solved with `numpy.linalg.solve`; $P_{in} = \text{Re}(V1 I_{in}^*)$, $P_{out} = |I2|^2 R_L$; inverter = SiC full bridge (C3M0021120K-class, 32 mOhm hot, ZVS turn-off 0.15 mJ scaled, 25 W control), rectifier = SiC Schottky bridge (0.9 V, 20 mOhm); V_{dc} found by bisection so that exactly 7.7 kW reaches the battery in every case.

■ Fields: $|B|_{rms}$ from all segments of both coils with the solved current phasors, validated against the on-axis loop formula (0.07 % error); air core only, no ferrite or aluminium shield, so the leakage field is an over-estimate.

4. Results (results/cases.csv, summary.json - 7.7 kW delivered in every case)

Energy-balance residual $P_{dc_in} - (P_{bat} + \text{all losses}) = 0.0$ W in every case. Coupling vs lateral offset at 175 mm: $k = 0.221 / 0.213 / 0.203 / 0.189 / 0.155 / 0.115$ at 0 / 50 / 75 / 100 / 150 / 200 mm. Field exposure with SS nominal currents (air core): 1,343 uT at the centre of the gap (inside the pad), 11.9 uT at the vehicle sill (0.80 m, 0.10 m) rising to 14.2 uT at +75 mm offset, 5.4 -> 6.3 uT at a person beside the car (1.00 m, 0.50 m) - compliant with ICNIRP 2010 (27 uT), marginal against ICNIRP 1998 (6.25 uT); the 27 uT contour sits at about $x = 0.65$ m.

Case	k	V _{dc}	I1 / I2 (A rms)	P _{inv}	P _{coil 1}	P _{Lf1}	P _{coil 2}	P _{rec t}	Link eta	Syste m eta
SS nominal (175 mm)	0.221	551 V	16.0 / 21.4	61 W	72 W	-	102 W	44 W	97.8 %	96.5 %
SS +75 mm offset	0.203	506 V	17.4 / 21.4	64 W	86 W	-	102 W	44 W	97.6 %	96.3 %
SS Z2 min (140 mm)	0.286	710 V	12.3 / 21.4	55 W	43 W	-	102 W	44 W	98.2 %	96.9 %
SS Z2 max (210 mm)	0.173	434 V	20.4 / 21.4	72 W	117 W	-	102 W	44 W	97.2 %	95.8 %
LCC-S nominal	0.221	404 V	16.0 / 21.4	75 W	69 W	24 W	102 W	44 W	97.5 %	96.1 %
LCC-S +75 mm offset	0.203	440 V	17.4 / 21.4	71 W	82 W	20 W	102 W	44 W	97.4 %	96.0 %
LCC-S Z2 min (140 mm)	0.286	313 V	12.3 / 21.4	96 W	41 W	40 W	102 W	44 W	97.7 %	96.0 %

KEY TAKEAWAY An analytical + filament-method model in place of COMSOL/Maxwell puts a 7.7 kW, 85 kHz SS link at 96.5 % DC-DC efficiency with $k = 0.22$ at 175 mm, losing only 0.2 % at +/-75 mm misalignment and staying under 27 uT beside the car.

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Case	k	V _{dc}	I ₁ / I ₂ (A rms)	P _{inv}	P _{coil 1}	P _{Lf1}	P _{coil 2}	P _{rect}	Link eta	System eta
LCC-S Z2 max (210 mm)	0.173	515 V	20.4 / 21.4	64 W	113 W	15 W	102 W	44 W	97.1 %	95.8 %
Compensation at 85 kHz			SS			LCC-S (400 V bus)				
C1 / ESR			16.32 nF / 57 mOhm			20.35 nF				
C2 / ESR			19.94 nF / 47 mOhm			19.94 nF				
L _{f1} / C _{f1}			-			42.5 uH / 82.5 nF				
Resonance check w ⁰ 2 L C			1.000 / 1.000			1.000 / 1.000 / 1.000				
R _L at 400 V, 7.7 kW vs R _{L,opt}			16.8 ohm vs 20.3 ohm			-				
V _{dc} over the Z2 range			434 -> 710 V (rises with k)			515 -> 313 V (falls with k)				

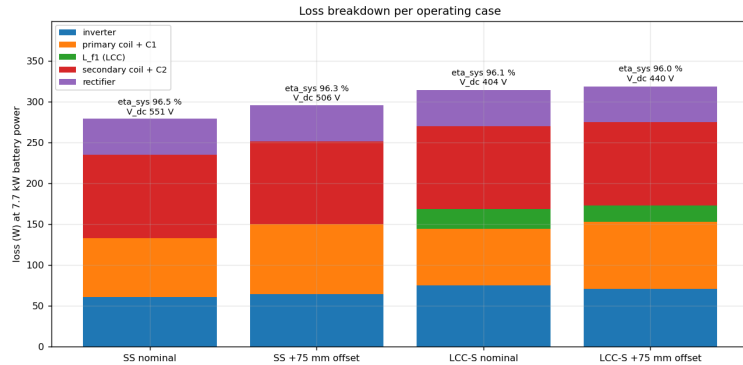


Fig. 2 - Loss breakdown per case: the secondary coil + C2 (102 W, fixed by the 21.4 A load current) is the largest item, then primary coil, inverter and rectifier.

5. Verification

Re-run here with the fixture-aware runner (pytest not installed): **15 passed, 0 failed** in 1.2 s.

Test	Checks	Result
test_mohan_matches_wheeler_published_formulas, test_mohan_matches_filament_method_within_10pct, test_mohan_against_published_example, test_single_loop_self_inductance_textbook_value	two published L formulas agree within 10 %; Mohan vs filament sum within 10 %; Grover loop 0.589 uH within 1 %	4 PASS
test_neumann_matches_maxwell_exact	filament M vs Maxwell elliptic-integral M within 1 % at three spacings	1 PASS
test_mutual_inductance_decreases_with_gap, test_mutual_inductance_decreases_with_misalignment, test_k_in_unit_interval	M monotone in gap and offset; $0 < k < 1$	3 PASS
test_ss_resonance_condition, test_lccs_resonance_conditions	$w_0^2 L C = 1$ to $1e-12$ for SS and all three LCC-S branches; $ I_1 = V_1/(w_0 L_{f1})$ load independent within 2 %	2 PASS
test_link_efficiency_in_unit_interval_and_increases_with_kQ, test_energy_balance_pin_equals_pout_plus_losses, test_vdc_solver_hits_target_power	eta in (0,1), monotone in k and kQ, numerical optimum load = analytical eta_max; $P_{in} = P_{out} + \text{losses}$ to $1e-9$; V _{dc} solver gives 7.7 kW to $1e-6$	3 PASS
test_biot_savart_matches_on_axis_loop_formula, test_field_decays_with_distance	Biot-Savart vs on-axis formula within 1 %; B decreases with distance	2 PASS

6. Data integrity and declared substitutions

■ **COMSOL / ANSYS Maxwell not used.** README section 8 maps each part to its FEM equivalent: Maxwell 3-D eddy-current solver at 85 kHz with litz conductor model, ferrite tiles and aluminium shield for L1, L2, M (inductance matrix) and Optimetrics sweeps; COMSOL Coil feature for litz; field overlays with a body phantom for ICNIRP. Nothing here ran in either tool.

■ Declared assumptions: ferrite factors 1.30 (L) and 1.40 (M) inside the range of Budhia, Covic and Boys (2011), absent from the field map; simplified Sullivan-type proximity factor and capacitor $\tan(\delta) = 5e-4$; fundamental-harmonic converters with ZVS assumed; device data from Wolfspeed datasheets; circular pads whereas J2954 reference pads are rectangular / DD; no thermal model, foreign-object detection, PFC loss or bifurcation analysis.

7. Learnings

SS needs the DC bus to rise with k (710 V at 140 mm) because its input impedance grows as $w_0^2 M^2 / R_L$, while LCC-S needs it to fall and gives a load-independent primary current - the reason J2954 reference designs use LCC. A +/-75 mm misalignment costs only 0.2 % efficiency because k drops just 8 % and the controller absorbs it through V_{dc}. Converter losses (105-120 W) and coil losses (170-200 W) are the same order, so coil Q matters as much as the semiconductor choice. Validating the filament solver against Maxwell's exact formula (< 1 %) is what gives me confidence in the k values without an FE tool.

KEY TAKEAWAY An analytical + filament-method model in place of COMSOL/Maxwell puts a 7.7 kW, 85 kHz SS link at 96.5 % DC-DC efficiency with $k = 0.22$ at 175 mm, losing only 0.2 % at +/-75 mm misalignment and staying under 27 uT beside the car.